ALLEGATIONS, NOT FINDINGS

Can pitch data catch *a rigged pitch?*

A pitcher-relative integrity monitor, tested on the 18 pitches federal prosecutors say were sold to bettors and on every MLB pitcher-month from 2023 to 2025.

pitch-integrity-monitor-uzwappgttfxqom7nyrx2nyk.streamlit.app
github.com/debi910/pitch-integrity-monitor

THE PROBLEM

A single pitch can now be bought.

- Pitch-level prop bets (ball or strike, speed, type) let a pitcher sell one pitch, not a game.
- Nov 2025: two Cleveland pitchers charged. Sept 2026 filing names 18 pitches; $400,000+ allegedly won.
- Every detection so far came from the betting side: someone noticed strange money.
- Nobody had measured whether the performance data itself gives a fixer away,
- or how many innocent pitchers a performance alarm would also flag.

OUR TWO QUESTIONS

1. Can public Statcast data reveal a fix?

2. What does the alarm cost in false alerts, league-wide?

THE DATA

Every MLB pitch, 2023–2025. All real.

2,175,365
pitches scored (2.18M pulled)

1,068
pitchers

8,339
pitcher-months

18 / 18
court-named pitches matched 1:1

SOURCE	WHAT	TERMS
Baseball Savant Statcast	via pybaseball, polite monthly pulls	MLBAM terms, research use
Sept 18 2026 EDNY filing (25-CR-346)	18 pitches: date, inning, batter, pitch #	Public record, allegations
DOJ press release, Nov 2025	charges, scheme description, Ortiz games	Public record

Nothing simulated or mocked. Raw pulls stay local; slim parquet files are in the repo.

THE METHOD

Judge each pitcher only against himself.

v1 · PER PITCH

Is this pitch strange for him?

Location, pitch choice, velocity

vs his own EARLIER games only

Fisher-combined p-value

Frozen in git 5fa3b3e before labels

v2 · PATTERN MONITOR

Has his rate jumped?

Template from the indictment's text:

first pitch, taken ball, in the dirt

Binomial tail vs his own baseline,

shrunk to the league; frozen 674637e

EVALUATION

Honest, league-wide

Simple baselines beside ours

False alarms on all 8,339 months

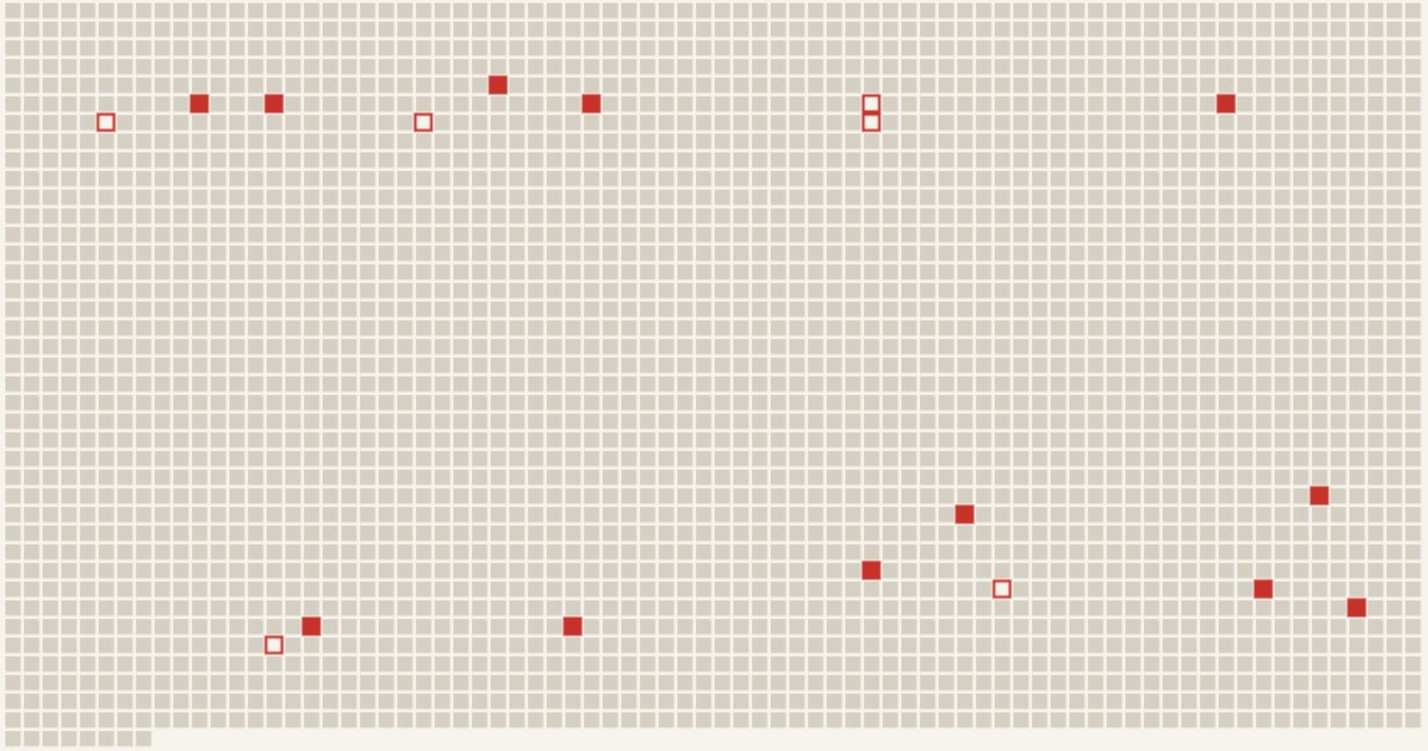
Stress test: 27 settings

Detection limit by injection

Code computes, nothing guesses: no LLM in the pipeline, every number reproducible with `make evals`.

RESULT 1 · SINGLE PITCHES

All 2,972 Clase pitches. 18 named.



model frozen in git · 5fa3b3e

AVERAGE PRECISION (higher is better)

Our model

0.10

One-line rule: spiked ball

0.20

Random

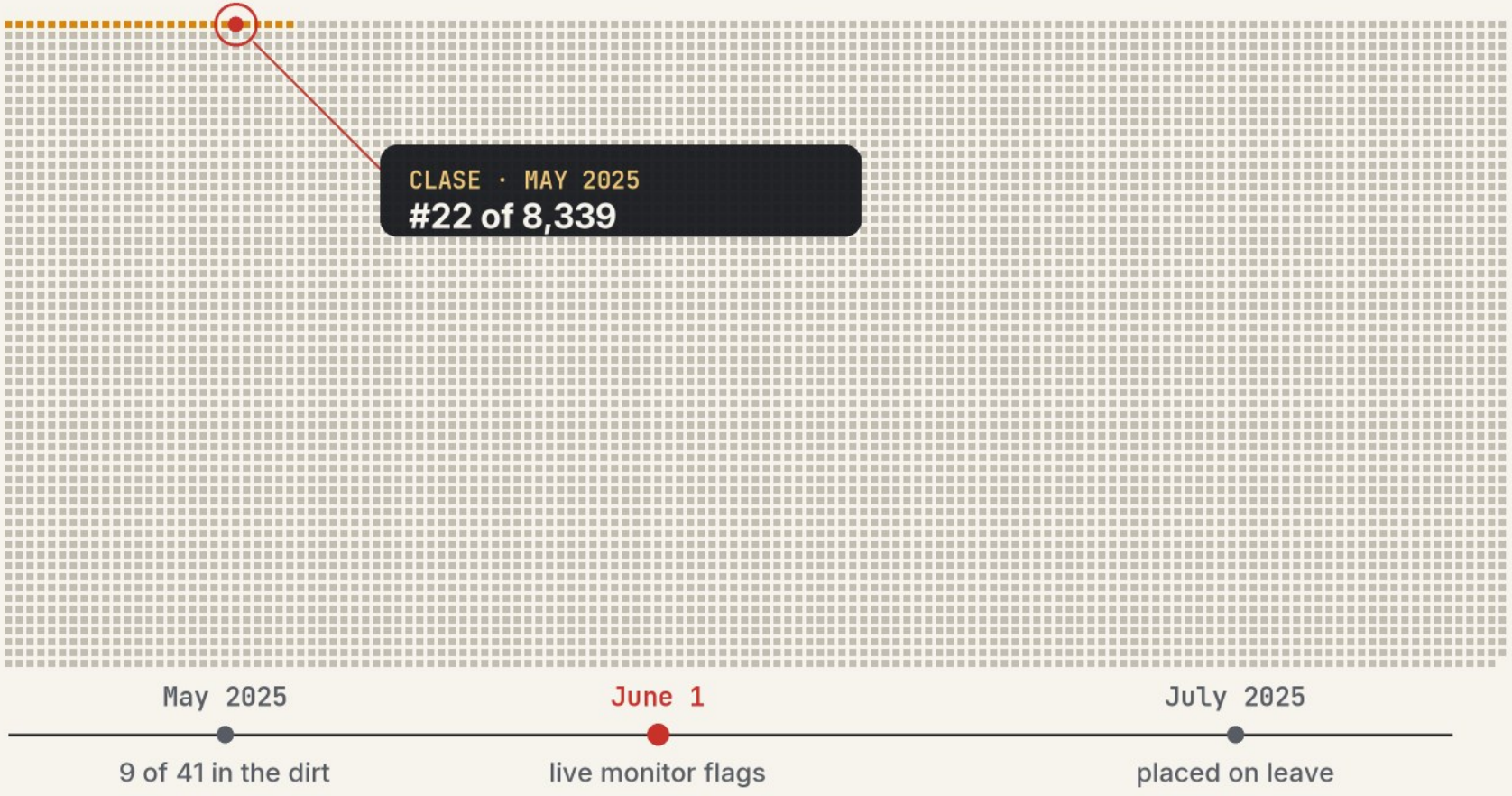
0.01

A rigged ball in the dirt
looks like an ordinary miss.

☐ 6 of 18 were not even balls

EVERY PITCHER-MONTH IN BASEBALL, 2023–25

8,339 pitcher-months, sorted by the monitor



RANK OF MAY 2025

Our monitor

#22

Raw monthly rate

#31

Raw monthly count

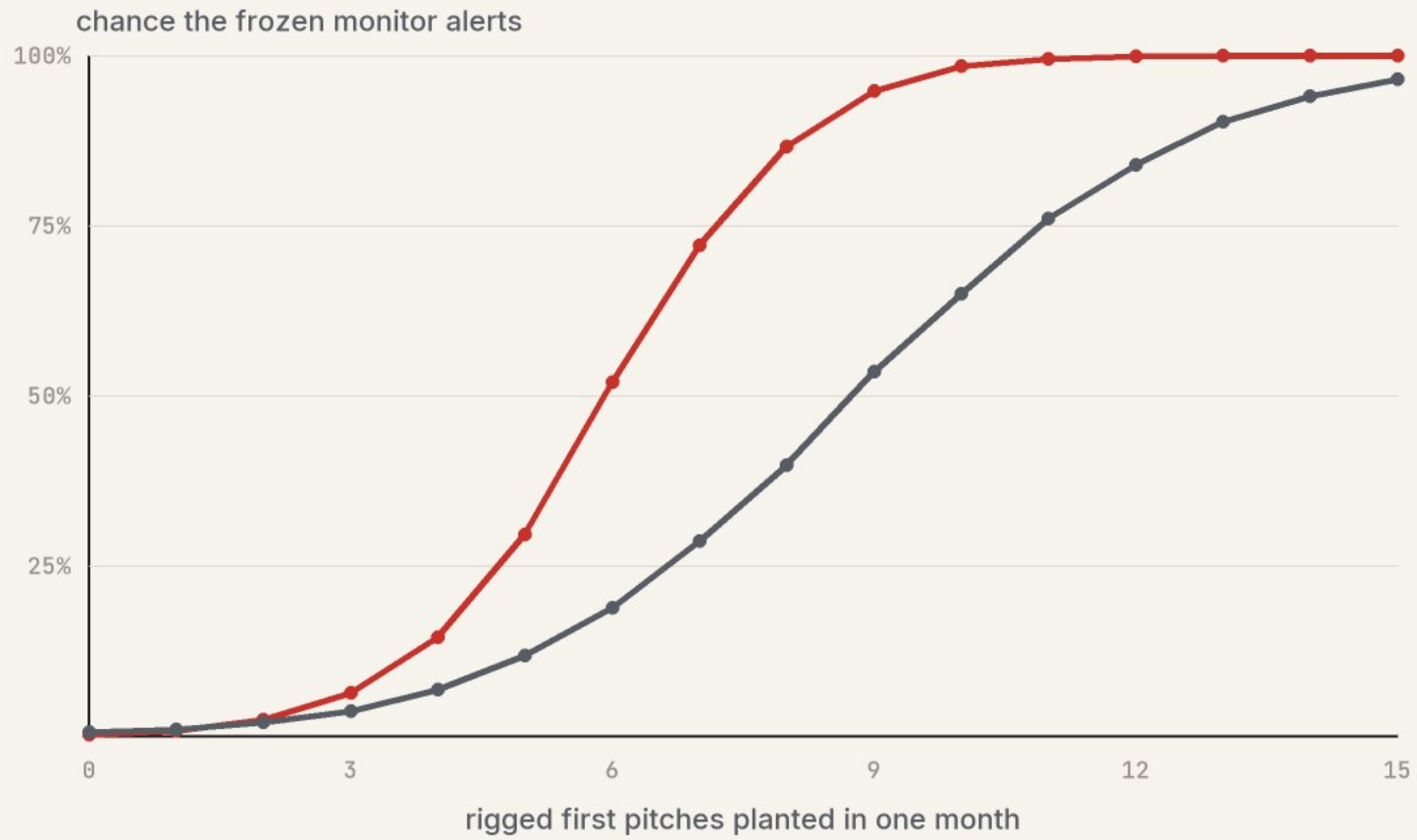
#783

1.08 alerts / month

27 in 3 seasons, whole league

DETECTION LIMIT

How many rigged pitches can the data see?



- **Relievers (20–59 first pitches / month)**
- **Starters (60+)**

~6 a month

for a 50% chance to catch a reliever

~9 a month

for a starter

Label-free: planted into real months, alert at the same ~1/month threshold.

WHO USES IT

A review list for an integrity analyst.

1 **Monthly run**
python -m pim season.csv

2 **Ranked list**
~1 alert a month, MLB-wide

3 **Analyst review**
pitches, video, betting data

4 **Decision**
humans, with betting evidence

USERS

- League integrity offices
- Sportsbook integrity desks
- Gaming regulators

NEXT

- Velocity and pitch-type monitors (other prop markets)
- Join with betting-volume data for a two-sided alert
- Minor leagues and other pitch-tracking sports
- Pilot with one integrity desk, measure analyst time

Ships as: live web app · CLI · Docker image · tested Python module · CI on every push

WHERE IT STOPS

Honest limits.

- It never outputs “rigged”. It ranks what is unusual for this pitcher.
- Single pitches: our frozen model lost to a one-line rule (AP 0.10 vs 0.20). Reported, not hidden.
- Caught 1 of 7 named months: sparse fixes do not move a monthly rate.
- Bets on speed or pitch type leave no trace in location.
- Labels are allegations; 18 positives make every interval wide.
- Other pitchers who trigger alerts are never named.

Both pitchers have pleaded not guilty; trial set for November 2026.